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Application of magnetron sputtering to deposit a multicomponent separator with polysulfide chemisorption and electrode stabilization for high-performance lithium-sulfur batteries

Jiayi Shi [a], Yingmei Yao [a], Liyuan Xue [a], Ke Li [a], Jinxia Ning [c], Feng Jiang [b], Fenglin Huang [a,*]

[a] College of Textile Science and Engineering, Jiangnan University, 1800 Lihu Avenue, Jiangsu Province, Wuxi 214122, PR China
[b] Department of Wood Science, University of British Columbia, Vancouver V6T 1Z4, Canada
[c] Jiangsu Anreda New Materials Limited Company, Changzhou 21300, China

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ABSTRACT

Lithium-sulfur battery has attracted immense attention because of its extraordinary high energy density and theoretical capacity. However, its practical application has been seriously hindered by the rapid capacity degradation associated with the "shuttle effect" and unstable electrodes. Such negative issues during charge-discharge process could be alleviated by modification of the separator, which is demonstrated in this paper. Here, we report a sandwich-type vanadium nitride/polyvinylidene fluoride-polymethylmethacrylate/hexagonal boron nitride separator (VN/PVDF-PMMA/hBN) by electrospinning of PVDF-PMMA nanofiber membrane, followed by magnetron sputtering of VN and hBN layer at the opposite size of the membrane. Compared to conventional coating method, magnetron sputtering led to an ultrathin coating layer of 200 nm. The sputtered VN/PVDF-PMMA/hBN separator showed higher porosity (68.7%) than commercial polypropylene (PP) separator (44.5%), and significantly high electrolyte uptake (335.8% s. 175.9%), which can lead to rapid transportation of lithium ions. In the meantime, the membrane showed clear benefit in inhibiting polysulfide diffusion and lithium metal electrode stabilization due to the presence of polar C=O groups that interact with the polysulfides. As a result, the Li—S battery with VN/PVDF-PMMA/hBN separator showed a high initial discharge capacity of 1077.4 mAh g^{-1} at 0.2C, which only slightly decreased to 905.1 mAh g^{-1} after 200 cycles. Significantly higher than the battery with commercial PP separator.

1. Introduction

The widespread application and development of miniaturized electronic equipment and large-scale electric vehicles have driven relentless pursuit of lithium secondary battery with high capacity and energy density as the energy storage systems [1,2]. Recently, due to the low cost, natural abundance and environmental friendliness of sulfur as well as the high theoretical specific capacity (1675 mAh g^{-1}) and high theoretical energy density (2600 Wh kg^{-1}) of lithium-sulfur battery approximately 3–5 times higher than that of traditional lithium-ion battery, lithium-sulfur battery has become a more promising candidate for next-generation rechargeable energy storage system [3,4]. Unfortunately, commercial application of lithium-sulfur battery is still limited due to a series of defects, including: (1) the insulating nature of sulfur and its reduction products of Li_2S_2 and Li_2S, which results in low utilization of sulfur active materials [5]; (2) the "shuttle effect" associated with high dissolution of polysulfide intermediates into the liquid electrolytes, which leads to low specific capacity and coulombic efficiency [6]; and (3) the serious growth of lithium dendrite in the anode, which brings potential security risks [7].

At present, there are many strategies to solve these crucial obstacles for high-performance lithium-sulfur batteries, such as directly modifying commercial separators, which are able to effectively suppress the dissolution of polysulfides intermediates during the charge and discharge for Li—S batteries [8,9]. However, polyolefin membranes with low porosity and poor affinity to electrolytes, which results in slow kinetics and poor rate capability during the charge and discharge process [10]. Electrospinning nanofiber membranes have been applied in batteries to improve porosity of separators attributing to their natural characteristic of high porosity and 3D porous structure, which greatly

* Corresponding author.
E-mail address: flhuang@jiangnan.edu.cn (F. Huang).

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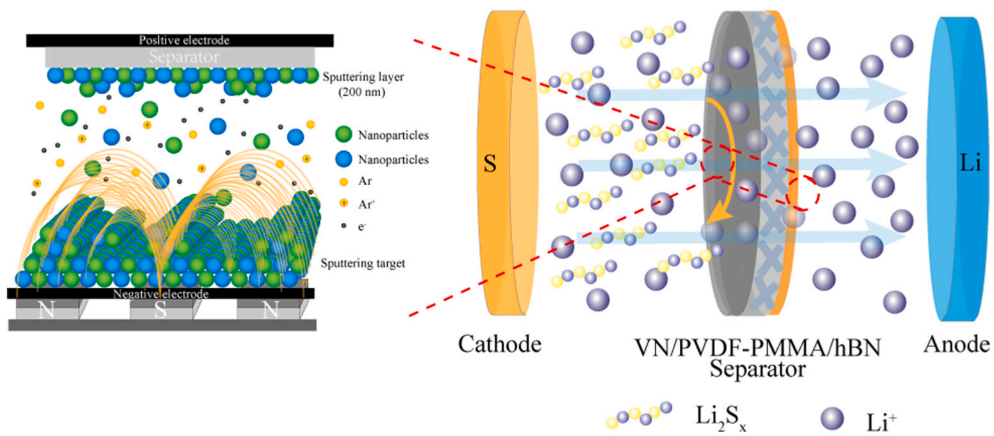

Fig. 1. Schematic of configuration of Li—S battery with the VN/PVDF-PMMA/hBN separator.

contributes excellent electrolyte uptake and increases ionic conductivity, and therefore are considered as promising separators for Li—S batteries. Although electrospun polyvinylidene fluoride (PVDF) nanofibers membrane contains high porosity and 3D porous structure, polysulfides cannot be effectively suppressed due to the low binding energy between the —F and Li_2S [11]. Innovatively, Cui et al. proposed that Li—S exhibits strong interaction with ester (-COO-) groups with the binding energy with Li_2S of 1.10 eV, which is much higher than the 0.40 eV between —F— and Li_2S [12]. Therefore, we hypothesized that by introducing polymethylmethacrylate (PMMA) with rich acrylate groups in the polyvinylidene fluoride (PVDF) polymer matrix, a porous PVDF-PMMA composite membrane with good mechanical properties and strong polysulfides immobilization effect can be designed.

Highly porous and conductive carbon materials, such as coating the Celgard polypropylene (PP) separator with graphene [13], multiwall carbon nanotubes [14] and carbon nanoparticles [15], have been used to improve the interfacial properties between electrode and separator, prevent the migration of polysulfides away from the cathode to the anode, and improve the cycling stability of the Li—S batteries. However, the "shuttle effect" cannot be effectively suppressed due to the weak interaction between the non-polar carbon surface and polar Li_2S_n. Most recently, polar inorganic oxi1des, such as silica [16], have been widely investigated and demonstrated to have strong chemical adsorption of polysulfide [17]. The insulating nature of metal oxides ultimately impedes electron transport and interrupts paths for Li ion movement, thus leading to low sulfur utilization and rate capability. On the other hand, metal nitrides, due to its high electrical conductivity and good affinity with polysulfide, have been identified as an ideal anchoring material for Li—S battery separator [18]. Sun et al. have developed a conductive porous vanadium nitride (VN) nanoribbon/graphene oxide (GO) composite separator, which has shown to lead to lower polarization, faster oxidation-reduction kinetics, superb rate and cycle performance for the Li—S battery [19].

Although great efforts in entrapping polysulfides on the cathode side to improve the cycle stability of the lithium-sulfur battery have been reported [20,21], a few polysulfides can still pass through the separator and cause lithium dendrite growth and deterioration on the anode side [22,23]. Uncontrolled Li dendrites growth and uneven heat distribution on the separator during charging and discharging process create potential risk of batteries safety [24]. Lithium anode corrosion can be reduced by inhibiting the polysulfide shuttle and the uniform deposition of lithium ions are promoted via hBN with conductivity of the anode side separator, which effectively improves the electrochemical performance of the lithium-sulfur battery [25–27]. High thermal conductivity can enable conformal thermal distribution of the separator upon repeated discharge-charge process, thus, contributing to the stable formation of solid electrolyte interface layer and uniform plating/striping of Li. Therefore, BN with high thermal conductivity (82 W mK^{-1}) and electrical conductivity has been used to modify separator for Li—S battery [26].

Herein, a multifunctional sandwich-type composite separator is prepared by electrospinning PVDF-PMMA into highly porous membrane. The surfaces of the membrane towards cathode and anode side of the Li—S battery were coated with VN and hBN via magnetron sputtering method, respectively (Fig. 1). Compared to conventional coating method, magnetron sputtering can lead to ultrathin coating layer of 200 nm, leading to smaller interfacial impedance. The VN/PVDF-PMMA/hBN separator with 3D interconnected network structure can accelerate the lithium ion diffusion and improve the chemical adsorption of polysulfide. The VN layer is not only used as current collector to promote the electron transfer, but also works as the first line of defense for suppressing polysulfide shuttle effect. The hBN layer can help to dissipate the heat generated during charging/discharging process, and can suppress the polysulfides penetration and reduced the growth of Li dendrites. Overall, the sandwich-type VN/PVDF-PMMA/hBN composite separator is expected to inhibit polysulfide shuttle effect, facilitate the electrons/ions transfer, and promote Li anode stability, all benefiting the electrochemical performance of Li—S battery.

2. Experimental

2.1. Fabrication of PVDF-PMMA nanofiber membrane

The PVDF-PMMA nanofiber membrane was synthesized by electrospinning. In brief, 3.5 g PVDF (PVDF, Mw = 800,000 g mol^{-1}) and 1.5 g PMMA (PMMA, Mw = 100,000 g mol^{-1}) were completely dissolved into 30 g DMF under strong magnetic stirring at room temperature for overnight to form a homogeneous solution. The PVDF-PMMA precursor solution was spun into nanofiber membranes by electrospinning at a flow rate of 0.8 mL/h and voltage of 16 kV using a needle with a diameter of 1 in. The nanofiber membrane was collected onto a grounded aluminum foil placed at 16 cm away from the syringe needle. After electrospinning, the prepared PVDF-PMMA membrane was dried in a vacuum oven at 60 °C for 8 h to remove remaining solvent.

2.2. Fabrication of VN/PVDF-PMMA/hBN composite separator

A sandwich-type VN/PVDF-PMMA/hBN membrane was prepared by magnetron sputtering of the PVDF-PMMA nanofiber membrane in the vacuum chamber (JZCK-420B, Shenyang Jingyi Research Technology Co Ltd., China). Pure V (99.99%, Hefei Crystal Materials Technology Co., Ltd.) and N_2 were used to prepare VN nanoparticles under the protection of Ar. The VN were uniformly deposited on the membrane (facing the cathode side) for 30 min with 150 W radio-frequency power

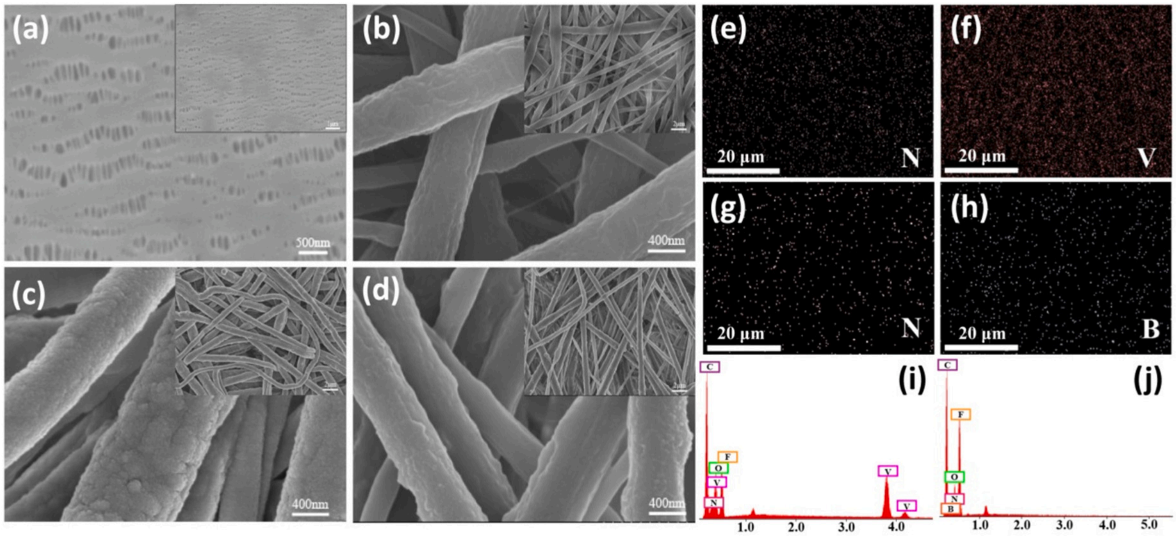

Fig. 2. SEM images of (a) PP, (b) PVDF-PMMA, and sandwich-type VN/PVDF-PMMA/hBN separator of (c) VN and (d) hBN coating sides. EDX (e–h) mapping and spectra (i & j) of VN/PVDF-PMMA/hBN separator of (e, f, i) VN and (g, h, j) hBN coating sides.

under safety pressure of 3 Pa. The hBN nanoparticles were deposited on the other side of the membrane (facing the anode side) by using hBN as a target under Ar atmosphere. The thickness of the each sputter-coated layer was controlled at 200 nm using a quartz crystal microbalance (FTM107-A, Shanghai Taiyao Co. Ltd., China) as a catcher. After deposition, the VN/PVDF-PMMA/hBN separators were cut into discs with diameter of 18 mm and dried in a vacuum oven of 50 °C.

2.3. Material characterization

The surface morphology of the separators was observed by a field-emissions scanning electron microscope (FE-SEM, S-4800, Hitachi, Japan). The temperature distribution images after removing the heating source were captured by an infrared (IR) camera (ICR, R500Ex-Pro-D) after illuminating to 40 °C by an infrared lamb. The thermal stability was examined from 30 to 200 °C at a 10 °C/min heating rate under N_2 by a differential scanning calorimetry (DSC, TA Q200).

The porosities of the separators were measured by comparing the volume of separators before and after soaking with n-butanol and calculated according to the following equation:

$$\mathrm{P}(\%) = \frac{M_{\mathrm{wet}}/\rho_b}{M_{\mathrm{wet}}/\rho_b + M_{\mathrm{dry}}/\rho_b} \times 100\% \tag{1}$$

where M_{dry} and M_{wet} are the weight of the dry and wet separators, respectively, ρ_b and ρ_p are the density of n-butanol and polymer, respectively.

The electrolyte uptake was investigated by soaking the separators in the liquid electrolyte (1 M LiTFSI and 0.1 M $LiNO_3$ in solvent mixture of 1, 3-dioxolane (DOL) and 1, 2-dimethoxyethance (DME) (1:1 v/v)) for 2 h at room temperature. Then the values of the electrolyte uptake were calculated according to the following equation:

$$Uptake(\%) = \frac{W - W_0}{W_0} \times 100\% \tag{2}$$

where W_0 and W are the weight of separator before and after absorbing the liquid electrolyte.

The contact angle was analyzed by investigating the electrolyte wettability of 25 μL electrolyte on the surface of separators using the optical contact angle measuring device (DCAT-21, GER Dataphysica Industry) at room temperature.

2.4. Sulfur cathode fabrication and battery assembly

The sulfur cathodes were fabricated by mixing 70 wt% of pure sulfur as active material, 20 wt% of carbon black as conductive materials and 10 wt% of PVDF binder in *N*-methyl-2-pyrrolidone (NMP) to from a homogeneous slurry. The slurry was pasted onto Al foil current collector and dried at 60 °C overnight under vacuum. Then, the areal loading of sulfur was around 2.37 mg cm^{-2} and used as the working electrodes. The coin-type Li—S battery were assembled in an Ar-filled glove box (including lithium metal foil as the anode, PP, PVDF-PMMA, VN/PVDF-PMMA/hBN as separators), and dropped the electrolyte of 1 M bis (trifluoromethane) sulfonamide lithium (LiTFSI) and 0.1 M $LiNO_3$ in a mixture of DOL and DME (1:1 by volume).

2.5. Electrochemical performance measurement

Diffusion coefficient of lithium ion (D_{Li+}) was evaluated via cyclic voltammetry measurement at various scan rates. The value of D_{Li+} was obtained by the Randles-Sevcik equation:

$$I_p = 2.69 \times 10^5 \mathrm{n}^{1.5} A D_{Li^+}^{0.5} C_{Li} v^{0.5} \tag{3}$$

where I_p is the peak current (A), n is the number of electrons involved in the reaction (n = 2 for Li—S battery), A is the area of electrode (cm^2), C_{Li} is Li^+ concentration in electrolyte (mol L^{-1}), and v is the CV scanning rate (V s^{-1}).

The galvanostatic charge-discharge properties of batteries were examined using the battery test equipment (CT-3008W-5V1nA-S4, Shenzhen Neware Technology Limited, China) within a potential range from 1.5 to 3 V by at different current densities. Cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) measurements were performed on a CHI 660 D electrochemical workstation (CH Instruments, China) in the frequency range of 10^5–10^{-2} Hz.

2.6. Polysulfide permeability evaluation

Li_2S_6 was synthesized by adding Li_2S and S at a molar ratio of 1:5 into an appropriate amount of DOL/DME (1:1 v/v) under an intensive magnetic stirring for 24 h at 45 °C until the sulfur fully dissolved. Especially, Li_2S_6 in DOL/DME was adjusted to 0.5 M. Then, different separators were installed in dialysis bottle with the 2 mL Li_2S_6 solution, and inverted in a device containing a transparent electrolyte. After 6 h, different colors of the electrolyte were observed due to the diffusion of

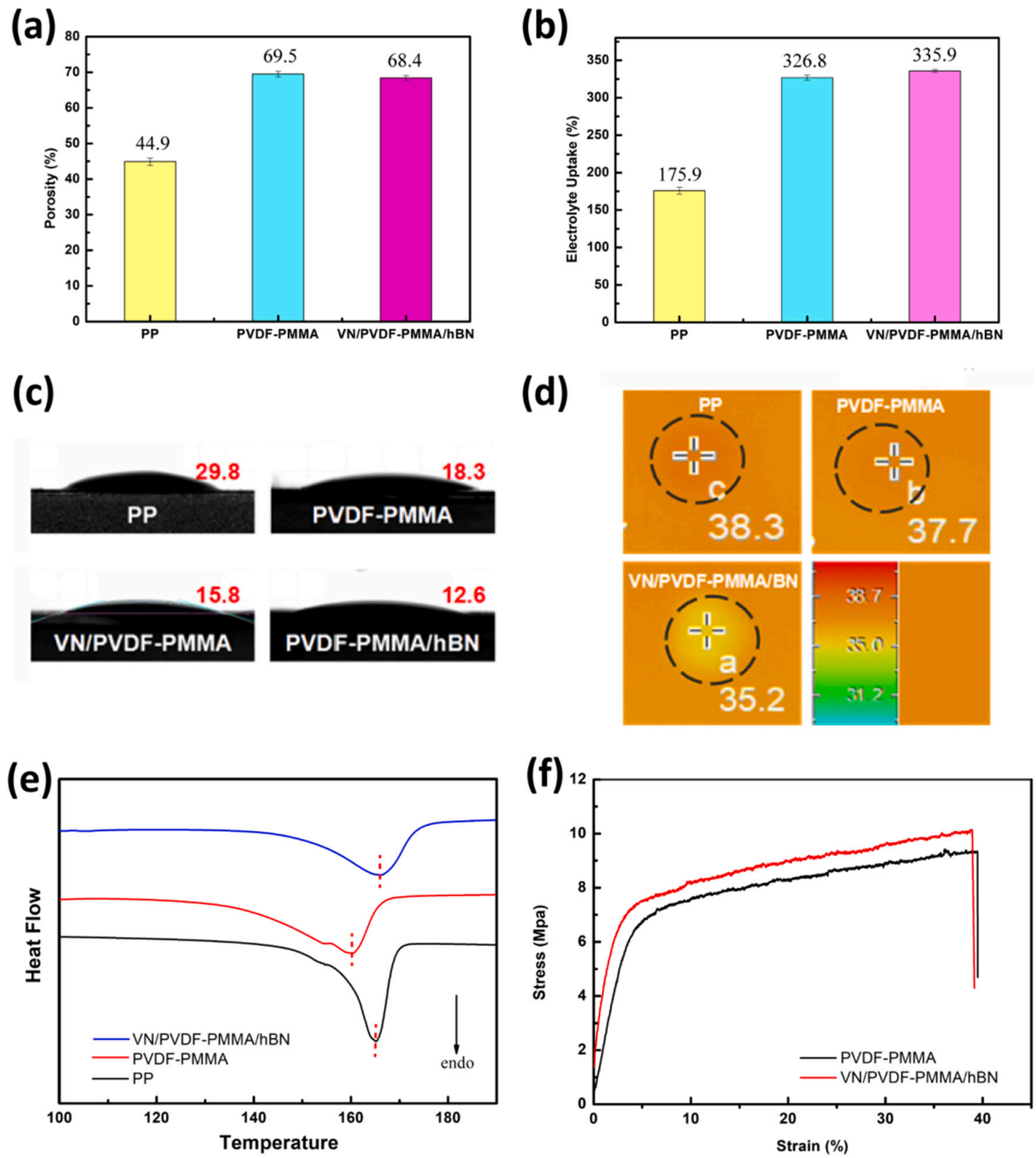

Fig. 3. (a) The porosity, (b) electrolyte uptake, (c) contact angles, (d) temperature distribution images, (e) DSC curves and (f) stress-strain curves of different separators.

Li_2S_6 caused by the concentration gradient difference.

3. Results and discussion

The surface morphologies of Celgard PP, electrospun PVDF-PMMA, and VN/PVDF-PMMA/hBN sandwich-type separators were investigated by scanning electron microscopy (SEM) (Fig. 2a–d). Commercial PP separator exhibits uniform slit-like submicron pores with length of around 200 nm (Fig. 2a). Such slit pores can be observed across the whole membrane and though the thickness direction. Although the open and through slit pores can help to maintain the ionic pathway and prevent direct contact between cathode and anode, they cannot inhibit polysulfides “shuttle effect” leading to severe capacity deterioration of lithium-sulfur batteries. In contrast, the electrospun PVDF-PMMA nanofiber membrane shows 3D interconnected network microstructure (Fig. 2b). Such interconnected microstructure is critical for maintaining high porosity with rapid transmission of lithium ions, while the increased tortuosity and the rich ester functional groups of the PVDF-PMMA can help to inhibit the “shuttle effect” of polysulfides [12]. Some aggregated morphologies on the SEM images of VN/PVDF-PMMA/hBN separator indicated successful and uniform deposition of VN and hBN nanoparticles on the two sides of sandwich-type VN/PVDF-PMMA/hBN separator, without affecting the overall porous structure (Fig. 2c & d). Energy Dispersive X-ray (EDX) mapping and spectra showed the presence of V and N atoms on one size of the separator, and B and N atoms on the opposite side (Fig. 2e–j).

Porosities and electrolyte uptakes of lithium-sulfur battery separator are critical for the battery performance, as were investigated for PP, PVDF-PMMA and VN/PVDF-PMMA/hBN separators (Fig. 3a & b). Compared to Celgard PP separator with porosity of 44.9%, the electrospun PVDF-PMMA membrane showed significantly higher porosity of 69.5% due to the 3D porous structure. Sputter coating of PVDF-PMMA

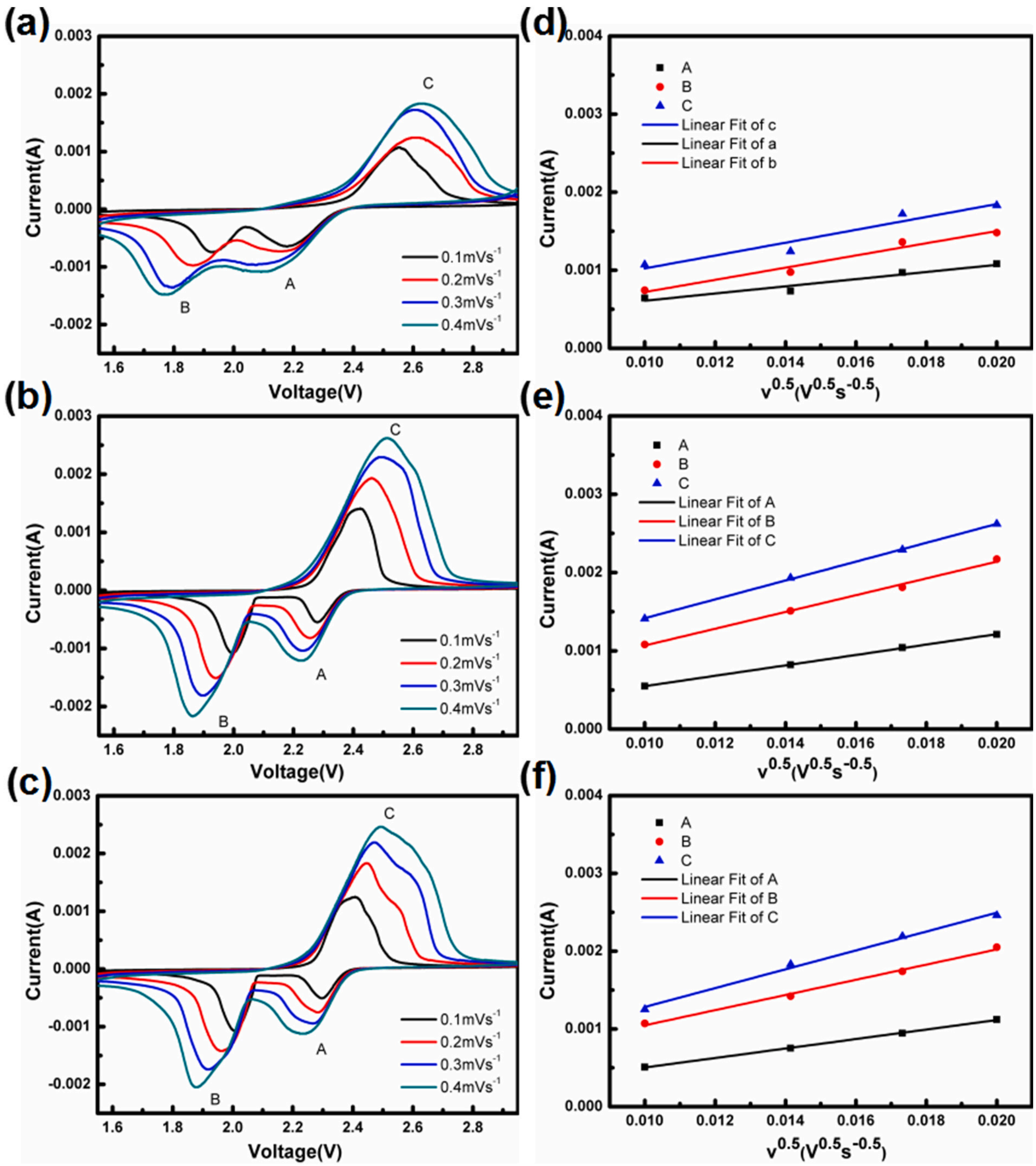

Fig. 4. Cyclic Voltammetry (CV) of Li—S batteries at various voltage scan rates and corresponding linear fits of the peak currents with (a, d) PP, (b, e) PVDF-PMMA and (c, f) VN/PVDFPMMA/hBN separators.

membrane with VN and hBN did not show significant effect on the porosity, which remains to be 68.4%. Correspondingly, both PVDF-PMMA and VN/PVDF-PMMA/hBN membranes showed high electrolyte uptake of 326.8 and 335.9%, respectively, much higher than the 175.9% for PP separator. The significant improvement of the porosity and the electrolyte uptakes for the PVDF-PMMA separator is attributed to the highly porous structure of the PVDF-PMMA nanofiber skeleton and stronger affinity between the electrolyte and polar C═O groups. Such high porosity and electrolyte uptake can ensure fast lithium ion transfer for excellent performances of Li—S batteries. Coating the PVDF-PMMA membrane with VN and hBN nanoparticle showed little effect on the porosity, while the electrolyte uptakes slightly increased as compared to the PVDF-PMMA membrane. This could be due to better electrolyte wettability and superior affinity of VN and hBN nanoparticles to the electrolyte. The wettability of electrolyte on the separator was further verified from the contact angles between the electrolyte and the separator (Fig. 3c). The VN/PVDF-PMMA/hBN separator exhibited contact angle of 15.8° and 12.6° on VN and hBN sides, respectively, smaller than the 29.8° and 18.3° for those of PP and PVDF-PMMA separator, suggesting higher affinity due to the presence of a large number of C═O groups and negative polar groups.

The separator plays a crucial role in the configuration of lithium sulfur battery to prevent direct contact between cathode and anode while allowing Li^+ transfer. In addition, due to the exothermal nature of Li deposition/striping during electrochemical reactions [24,26], the separator should be able to dissipate the heat and maintain good thermal

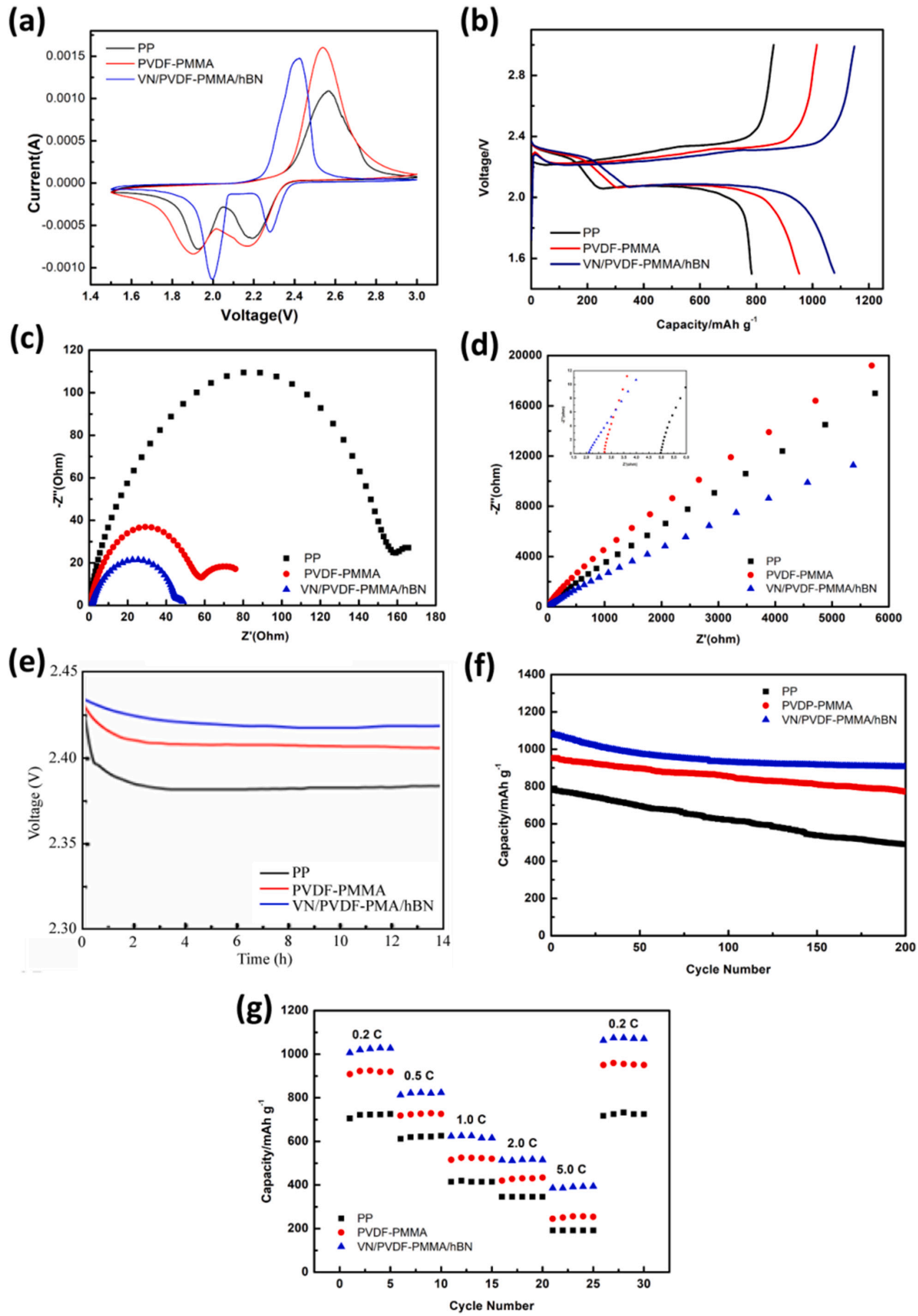

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Fig. 5. Electrochemical performance of Li—S batteries PP, PVDF-PMMA and VN/PVDF-PMMA/hBN separators. (a) CV curves of Li—S batteries at a scan rate of 0.1 mV s^{-1}, (b) open circuit voltage profiles showing self-discharge behavior, (c) Nyquist plots of Li—S batteries before cycling, (d) AC impedance spectra, (e) Evolution of open-circuit voltages with time, (f) cycling performance of Li—S batteries at 0.2C (1C = 1675 mA g^{-1}), (g) rate performance of Li—S batteries at current densities from 0.2 to 5C.

stability to avoid internal short circuit caused by the contact between cathode and anode. Heat dissipation of different separators after illuminating with infrared lamb at 40 °C is presented in Fig. 3d. Upon removing the infrared lamb, both PP and PVDF-PMMA separators presented a hotspot with temperature of 38.3 and 37.7 °C, respectively. However, with the deposition of hBN, the temperature of VN/PVDF-PMMA/hBN separator significantly reduced to 35.2 °C, as hBN nanoparticles with higher thermal conductivity permits localized heating to spread more easily. It can be seen that the hBN layer can create a more uniform thermal distribution, which would facilitate more uniform growth/dissolution of Li and thus better cycling stability and rate performance [25]. Other than heat dissipation, thermal stability is also important in preventing membrane melting or shrinking at high temperature. The thermal stability of the PVDF-PMMA membrane was characterized using DSC (Fig. 3e), showing endothermic peak due to polymer melting. The melting temperature of electrospun PVDF-PMMA is around 160.9 °C, which slightly increased to 165.9 °C for the VN/PVDF-PMMA/hBN separator due to the presence of inorganic nanoparticles coating. Nevertheless, the melting temperature of both membranes is significantly higher than the safe operating temperature [28] of battery separator, suggesting that they could be used for high temperature battery applications. The mechanical properties of separator play an important role in the performance and working life of a battery. In this work, tensile tests of uncoated and VN//hBN-coated separators were performed at room temperature. The typical stress-strain curves are showed in Fig. 3f. It can be seen that the tensile strength of the sputtered separator exhibited slightly higher than that of the original separator. This phenomenon may be due to the fact that sputtered ions can compensate for the surface flaws of nanofibers during manufacturing, thereby improving the strength of the fibrous separator [29,30].

The Li-ion diffusion coefficient (D_{Li+}) of different separators was detected with a series of cyclic voltammetry (CV) measurements at different scan rates (Fig. 4a–c). Randles–Sevick equation [31,32] was used for theoretical calculation. Herein, characteristic cathodic peaks are assigned at around 2.2 V and 1.95 V, which are labeled as A and B, indicating the reduction of elemental sulfur to long-order polysulfides (Li_2S_x, x = 4, 6, 8) and long-order polysulfides to short-order polysulfides (Li_2S_2/Li_2S), respectively. The anodic peak at about 2.45 V is defined as peak C, corresponding to conversion of insoluble Li_2S_2/Li_2S to soluble polysulfides and further to sulfur. According to Eq. (3), the plot of the reduction peak current I_p versus the square root of scan rate could be linearly fitted as shown in Fig. 4d–f, and the slope of the linear fitting curves represents the lithium diffusion coefficients (D_{Li+}) of the different separators. The lithium diffusion coefficients were determined as D_{Li+} (A) = 2.90×10^{-9} cm^2 s^{-1}, D_{Li+} (B) = 6.36×10^{-9} cm^2 s^{-1}, and D_{Li+} (C) = 1.07×10^{-8} cm^2 s^{-1} for the Li—S battery with PP separator, which increased to D_{Li+} (A) = 5.51×10^{-9} cm^2 s^{-1}, D_{Li+} (B) = 6.71×10^{-9} cm^2 s^{-1}, D_{Li+}(C) = 1.65×10^{-8} cm^2 s^{-1} with PVDF-PMMA separator due to the dramatic increase in electrolyte uptake by the highly porous structure. After sputter coating, the diffusion coefficient of the VN/PVDF-PMMA/hBN separator slightly reduced to D_{Li+} (A) = 4.83×10^{-9} cm^2 s^{-1}, D_{Li+}(B) = 6.56×10^{-9} cm^2 s^{-1}, and D_{Li+} (C) = 1.34×10^{-8} cm^2 s^{-1}.The slight decrease in the diffusion coefficient is caused by the sputtered layer, but it is still higher than that of the PP separator. Obviously, the PVDF-PMMA and VN/PVDF-PMMA/hBN separators showed higher lithium-ion diffusion coefficients, which can promote rate performance of batteries.

The cyclic voltammetry curves were conducted to investigate the electrochemical behavior of Li—S batteries at a scan rate of 0.1 mV s^{-1}. As shown in Fig. 5a, the CV curves of the battery with the VN/PVDF-PMMA/hBN separator exhibited the sharp cathodic peaks located at around 2.0 and 2.3 V and two overlapped anodic peaks at about 2.4 V. In addition, the Li—S battery with the VN/PVDF-PMMA/hBN separator exhibits higher reduction potential and lower oxidation potential demonstrating significantly promoted redox kinetics of polysulfide for the battery, which achieve rapid polysulfide conversion during discharge–charge process. The discharge–charge profiles of the initial cycle for Li—S batteries with PP, PVDF-PMMA and VN/PVDF-PMMA/hBN separators were presented at a current density of 0.2C, as shown in Fig. 5b. In general, the typical discharge voltage profile is observed in the reduction process of Li—S battery, which are in the range of 2.4–2.3 V (the first plateau) and 2.3–1.9 V (the second plateau). The first plateau represents the transformations of elemental sulfur to long-chain polysulfides (Li_2S_x, $4 \leq x \leq 8$) and the second discharge plateau indicates the conversion of Li_2S_x to solid-phase low-order Li_2S_2 or Li_2S [33–35]. During the charging processes of Li—S battery, the two continuous charge plateaus correspond to the oxidation reactions from lithium sulfides to polysulfides and finally to sulfur. Therefore, the plateaus for the charge–discharge precisely corresponded to the peak positions of the CV curves, further to confirm redox reactions of sulfur and polysulfides. Fig. 5b indicated that the highest capacity for battery with VN/PVDFPMMA/BN separator, indicating more efficient kinetic reaction process due to its good wetting and electrolyte uptake properties.

Electrochemical impedance spectroscopy is conducted to evaluate the interfacial resistance between the separator and the electrode. The Nyquist plots of various Li—S batteries with PP, PVDF-PMMA and VN/PVDF-PMMA/hBN separators before cycling were exhibited in Fig. 5c, with the diameter of the semicircle in high frequencies representing the charge transfer resistance (R_{ct}). It is obvious that the interfacial resistance values of the lithium-sulfur battery with the VN/PVDF-PMMA/hBN separator (42.8 Ω) is smallest than that of bare PP separator (158.4 Ω) and PVDF-PMMA separator (59.2 Ω). The significant decrease in the R_{ct} for VN/PVDF-PMMA/hBN separator is due to its strong affinity to the electrolyte and the high conductivity of coated VN layer (2.1 S cm^{-1}). In addition, with 200 nm coated layer, sputter-coating barely increased the thickness of the separator. Fig. 5d shows the ionic conductivity of different separators with PP, PVDF-PMMA and VN/PVDF-PMMA/hBN, where the intercept of the curve and the horizontal axis represents the bulk resistance of the separator. It is worth noting that the excellent electrolyte uptake makes the VN/PVDF-PMMA/hBN membrane has a high ionic conductivity of 1.783 mS cm^{-1} compared to the PP membrane (0.246 mS cm^{-1}), which ensures fast lithium ion transfer for high performances of Li—S batteries. Self-discharge phenomenon was also conducted by monitoring open circuit voltage (OCV) of the Li—S batteries with PP, PVDF-PMMA and VN/PVDF-PMMA/hBN separators (Fig. 5e). Within a few hours, a drastic decay in the OCV (2.38 V) of the battery with Celgard PP separator occurs indicative of severe self-discharge. On the contrary, self-discharge is efficiently depressed by using the VN/PVDF-PMMA/hBN separator, showing a stable OCV of 2.42 V.

Fig. 5f presented cycling performance of the Li—S batteries with PP, PVDF-PMMA and VN/PVDF-PMMA/hBN separators at a current density of 0.2C. The initial discharge capacities of Li—S battery with PP, PVDF-PMMA, and VN/PVDF-PMMA/hBN were measured as 783.4, 952.7 and 1077.4 mAh g^{-1}, which further decreased to 490.6, 773.9, and 905.1 mAh g^{-1} after 200 cycles, respectively. The Li—S battery with PP separator showed the smallest and most severe decay (37%) in discharge capacity, which is due to the shuttling effect of the polysulfide and is consistent with previous observation [33,36,37]. Much higher discharge

Table 1
Comparison of the electrochemical performance of VN/PVDF-PMMA/hBN separator with other modified separators in literature.

Separator	Sulfur content in the cathode (%)	Sulfur mass loading (mg/cm^2)	Thickness of coating (μm)	Capacity after cycling (mAh/g)	Cycles (times)	Decay rate per cycle (%)	Reference
PAA separator	65	2.2		783.5 at 0.2C	200	0.26	[33]
TiN coated separator	78.14	1.7	21.8	605 at 0.5C	250	0.17	[39]
ACNF interlayer	70	2.5		897 at 0.1C	100	0.27	[35]
MXene coated separator	70	2.07	25.0	876.9 at 0.5C	250	0.22	[38]
rGO/PEOF:PSS separator	70	1.0	20.0	800 at 0.5C	100	0.24	[37]
Uio-66-5/Nafion coated separator	80	1.7	4.0	851.1 at 0.1C	200	0.032	[36]
VN/PVDF-PMMA/hBN separator	70	2.37	0.2	905.1 at 0.2C	200	0.079	The work

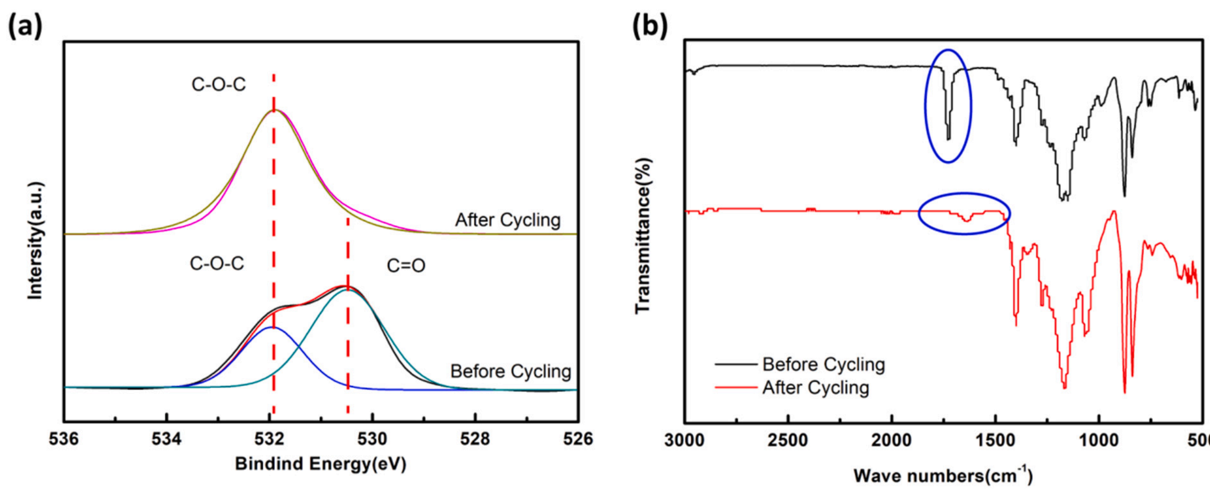

Fig. 6. (a) XPS spectrum at O1s and (b) FTIR spectra of pristine and cycled PVDF-PMMA separators.

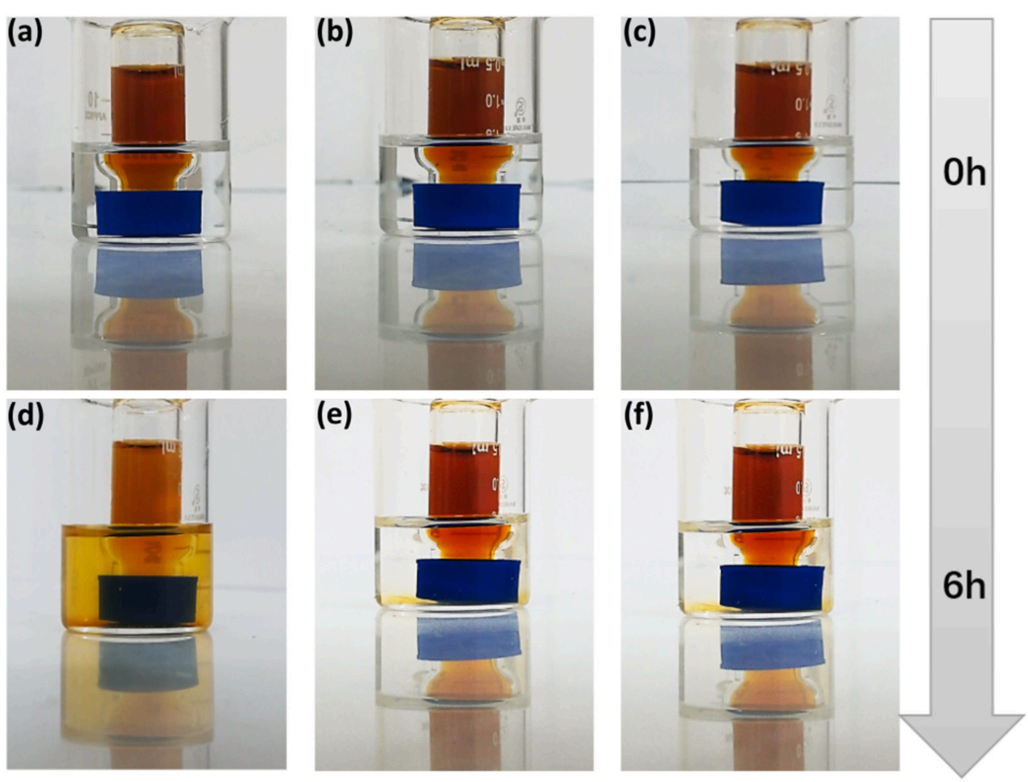

Fig. 7. The images of polysulfide diffusion across of (a, d) PP, (b, e) PVDF-PMMA and (c, f) VN/PVDF-PMMA/hBN separators in 6 h.

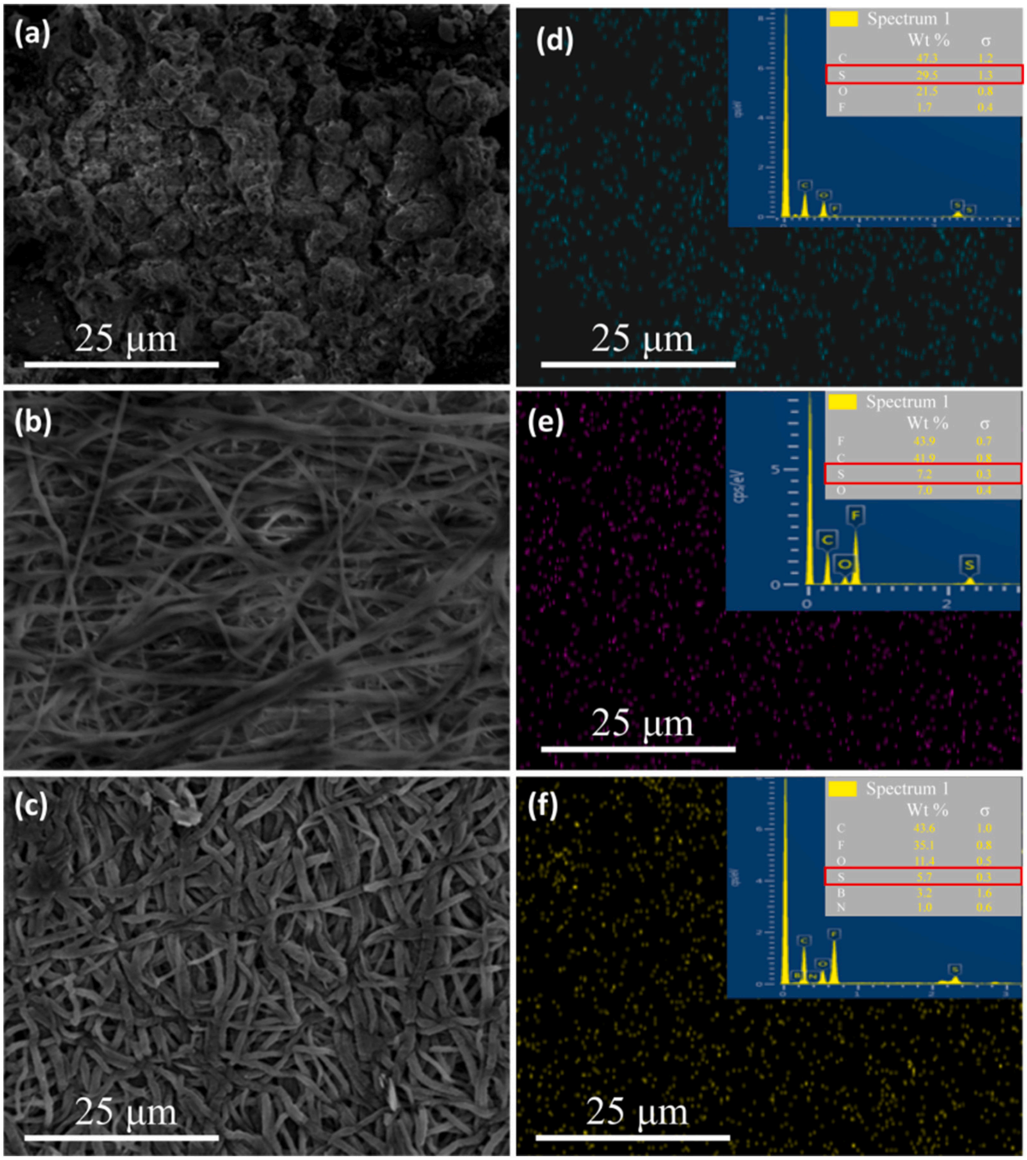

Fig. 8. (a–d) SEM images and (d–f) corresponding EDX sulfur mapping of (a, d) PP, (b, e) PVDF-PMMA and (c, f) VN/PVDFPMMA/hBN separators towards the Li anode side after 200 cycles at 0.2C.

capacity and less capacity decay could be observed for both PVDF-PMMA (19% capacity decay) and VN/PVDF-PMMA/hBN (15% capacity decay) separators, which are believed to be due to the relatively higher binding energies between the C═O and Li_2S/polysulfides compared with those between -CH_3 and Li_2S/polysulfides, resulting in suppressed polysulfide diffusion rate and improved active material utilization [38]. The rate capability of Li—S batteries with different separators at various rates was shown in Fig. 5g. The Li—S battery with VN/PVDF-PMMA/hBN separator deliver specific capacities of 1027.7, 824.0, 615.3, 515.3, 394.1 mAh g^{-1} at the rate of 0.2, 0.5, 1.0, 2.0, 5.0C, respectively, which are high capacities of the Li—S battery with VN/PVDF-PMMA/hBN separator returned to 1070.7 mAh g^{-1}, confirming a superior rate performance. The significantly higher specific capacity and excellent rate performance of Li—S battery with VN/PVDF-PMMA/hBN separator can be ascribed to the strong interaction between the polar ester functional group and Li_2S_n, as well as the high conductivity of VN coating layer, which can suppress the diffusion of polysulfides and improves the utilization of the active material. The electrochemical performance of the Li—S battery with the VN/PVDF-PMMA/hBN separator was compared with modified separators with different coating thicknesses (Table 1).er than the batteries with PP and PVDF-PMMA separators. When the current density is back to 0.2C, the specific.

In previous reports, strong interaction between C═O groups and polysulfides has been confirmed from first principle calculation with density functional theory [2]. As shown in Fig. 6, the XPS and FTIR spectrums of original and cycled PVDF-PMMA separator were further characterized to investigate the chemical interactions between C═O groups and polysulfides. The O 1s XPS spectra (Fig. 6a) showed original PVDF-PMMA separator with two different peaks located at 531.9 eV, 530.5 eV, corresponding to the typical C-O-C bond and C═O bond in PVDF-PMMA polymer matrix composite separator, respectively [40]. Noticeably, the C═O peak in the O 1s spectra of the PVDF-PMMA separators was barely captured after 200 cycles, which is attributed to lithium polysulfide directly bonding with the double bond oxygen atom in C═O, forming Li—O and covering most of the surface signal of the ester group [12]. Fig. 6b shows the FTIR spectra of PVDF-PMMA separator before and after 200 cycles. The pristine PVDF-PMMA separator showed a noticeable peak at 1728.8 cm^{-1}, which is assigned to the stretching vibration of C═O bonds. After 200 cycles, The formation of C-O…Li_2S_X between C═O and lithium polysulfide caused the dielectron

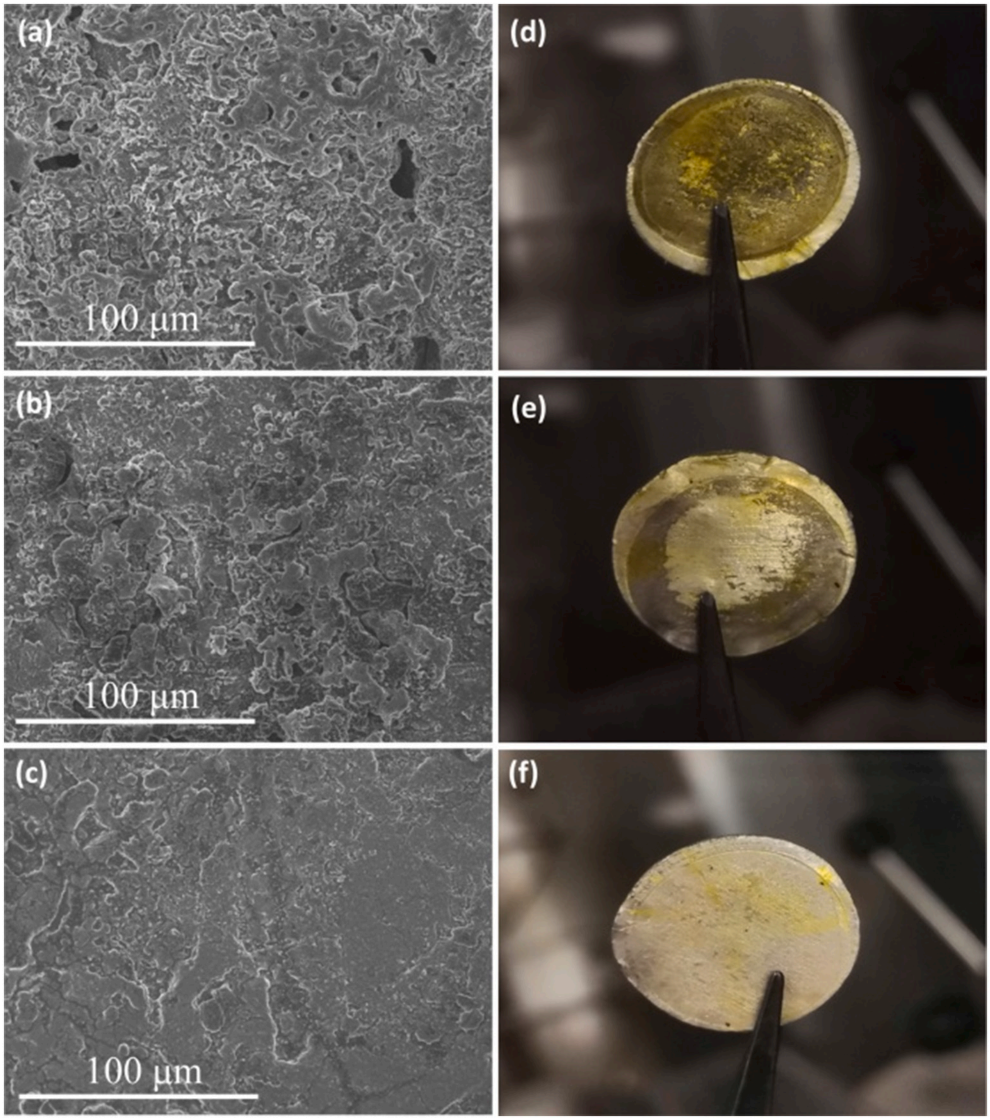

Fig. 9. Morphologies and photographs of cycled lithium metal after 200 cycles at 0.2C in Li—S batteries with (a, d) PP, (b, e) PVDF-PMMA, and (c, f) VN/PVDF-PMMA/hBN separators.

characteristic of C═O to weaken, and the corresponding peak moved to a low wave number (1630.0 cm^{-1}) and almost disappears, indicating that the ester group on the PVDF-PMMA membrane chemically anchors the intermediate polysulfide. Therefore, both XPS and FTIR spectra indicated strong interaction between the PVDF-PMMA separator and polysulfide, which may alleviate shuttle effect and protect metallic Li from polysulfides, thus improving the cycling stability.

To further demonstrate the inhibition of polysulfides by various separators, a simple self-made diffusion device was prepared to simulate the diffusion and penetration of polysulfide through PP, PVDF-PMMA and VN/PVDF-PMMA/hBN separators (Fig. 7). The polysulfides are completely blocked by all three separators at the beginning. After 6 h, the originally colorless electrolyte with PP separator turns from colorless to yellow. In contrast, the electrolyte solutions remained colorless for the beakers with PVDF-PMMA and VN/PVDF-PMMA/hBN separators, suggesting polysulfide migration could be suppressed by the highly porous and polar PVDF-PMMA membranes. Such polysulfide diffusion inhibitory effect can prevent shuttle effect and promote better electrochemical performance of the lithium-sulfur batteries, as shown in previous electrochemical performance results.

To further confirm the effectiveness of suppressed polysulfide diffusion, the surface morphologies and chemical composition of the separators in Li—S batteries towards Li anode side after 200 cycles at 0.2C current were investigated (Fig. 8). Severe large aggregates could be observed on the PP separator after 200 cycles (Fig. 8a), suggesting the pores are largely clogged by polysulfides, which can block the lithium ions transfer. In contrast, highly porous and fibrous morphologies could be observed for both PVDF-PMMA and VN/PVDF-PMMA/hBN separators (Fig. 8b & c), indicating the presence of open pores for lithium ion transfer. Furthermore, VN/PVDF-PMMA/hBN separator map a weaker S concentration among three separators (Fig. 8d, e & f), which may be derived from the relatively high adsorption capacity between the sulfur-related species and multicomponent sandwich-type composite separator.

The lithium metal anodes after 200 cycles at 0.2C were investigated to further confirm the polysulfide diffusion through the separators (Fig. 9). The Li metal anode in the Li—S battery with PP separator showed severe formation of Li dendrite (Fig. 9a) and appeared as yellowish color due to the contamination by insoluble polysulfides (Li_2S_2/Li_2S) (Fig. 9d), which can accelerate battery failure [41]. In contrast, the Li metal anode became cleaner and smoother for Li—S batteries with PVDF-PMMA separators, and appeared as less yellowish color due to the reduced accumulation of Li_2S_2 and Li_2S species on the surface of the Li metal. Both the separator side and lithium metal anode side confirmed that the polysulfide shuttle effect can be significantly suppressed by the VN/PVDF-PMMA/hBN separator due to the strong chemical interaction between ester functional group and lithium polysulfides, which can guarantee better electrochemical performance.

4. Conclusions

In summary, a VN/PVDF-PMMA/hBN separator was developed by electrospinning and magnetron sputtering and used as separators for highly stable Li—S batteries. The highly porous and polar VN/PVDF-PMMA/hBN separator presented several synergistic benefits of chemisorbing polysulfides and stabilizing electrodes, including, (1) the PVDF-PMMA nanofiber membrane not only improves electrolyte affinity, but also promotes Li^+ migration; (2) the VN layer with high conductivity can serve as the upper current collector to offer an additional electron pathway on the cathode side; (3) The BN layer possessed high thermal conductivity to ensure the stability and safety on the anode side; (4) components such as PMMA, VN and hBN with adequate surface adsorption sites and functional groups can play a key role in adsorbing soluble intermediates and suppressing the shuttle effect of polysulfide. Thus, we have demonstrated a strategy for developing VN/PVDF-PMMA/hBN separator with excellent electrochemical performance such as high specific capacity, superior rate performance and long-cycle life for high-performance lithium-sulfur battery.

CRediT authorship contribution statement

Jiayi Shi: Conceptualization, Methodology, Investigation, Writing - review & editing. **Yingmei Yao:** Methodology, Validation. **Liyuan Xue:** Software, Data curation. **Ke Li:** Resources, Formal analysis. **Jinxia Ning:** Writing - original draft, Visualization. **Feng Jiang:** Writing - review & editing. **Fenglin Huang:** Supervision, Writing - review & editing, Funding acquisition.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled "Application of magnetron sputtering to deposit a multicomponent separator with polysulfide chemisorption and electrode stabilization for high-performance lithium-sulfur batteries".

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References

[1] N. Deng, W. Kang, Y. Liu, J. Ju, D. Wu, L. Li, B.S. Hassan, B. Cheng, A review on separators for lithium sulfur battery: progress and prospects, J. Power Sources 331 (2016) 132–155.
[2] Z.W. Seh, Y. Sun, Q. Zhang, Y. Cui, Designing high-energy lithium-sulfur batteries, Chem. Soc. Rev. 45 (2016) 5605–5634.
[3] Y. He, Y. Qiao, H. Zhou, Recent advances in functional modification of separators in lithium-sulfur batteries, Dalton Trans. 47 (2018) 6881–6887.
[4] J.-Q. Huang, Q. Zhang, F. Wei, Multi-functional separator/interlayer system for high-stable lithium-sulfur batteries: progress and prospects, Energy Storage Mater. 1 (2015) 127–145.
[5] L. Fan, M. Li, X. Li, W. Xiao, Z. Chen, J. Lu, Interlayer material selection for lithium-sulfur batteries, Joule 3 (2019) 361–386.
[6] Y. Diao, K. Xie, S. Xiong, X. Hong, Shuttle phenomenon-the irreversible oxidation mechanism of sulfur active material in Li–S battery, J. Power Sources 235 (2013) 181–186.
[7] L. Zhang, Y. Wang, Z. Niu, J. Chen, Advanced nanostructured carbon-based materials for rechargeable lithium-sulfur batteries, Carbon 141 (2019) 400–416.
[8] X. Yu, J. Joseph, A. Manthiram, Suppression of the polysulfide-shuttle behavior in Li–S batteries through the development of a facile functional group on the polypropylene separator, Mater. Horiz. 3 (2016) 314–319.
[9] J. Zhu, Y. Ge, D. Kim, Y. Lu, C. Chen, M. Jiang, X. Zhang, A novel separator coated by carbon for achieving exceptional high performance lithium-sulfur batteries, Nano Energy 20 (2016) 176–184.
[10] J.H. Kim, G.Y. Jung, Y.H. Lee, J.H. Kim, S.Y. Lee, S.K. Kwak, S.Y. Lee, Polysulfide-breathing/dual-conductive, heterolayered battery separator membranes based on 0D/1D mingled nanomaterial composite mats, Nano Lett. 17 (2017) 2220–2228.
[11] J.Z. Pei Zhu, Jun Zang, Chen Chen, Lu Yao, Mengjin Jiang, Chaoyi Yan, Mahmut Dirican, Ramakrishnan Kalai Selvan, Xiangwu Zhang, A novel bi-functional double-layer rGO–PVDF/PVDF composite nanofiber membrane separator with enhanced thermal stability and effective polysulfide inhibition for high-performance lithium–sulfur batteries, J. Mater. Chem. A 5 (2017) 15096–15104.
[12] Z.W. Seh, Q. Zhang, W. Li, G. Zheng, H. Yao, Y. Cui, Stable cycling of lithium sulfide cathodes through strong affinity with a bifunctional binder, Chem. Sci. 4 (2013).
[13] L. Kong, B.-Q. Li, H.-J. Peng, R. Zhang, J. Xie, J.-Q. Huang, Q. Zhang, Porphyrin-derived graphene-based nanosheets enabling strong polysulfide chemisorption and rapid kinetics in lithium-sulfur batteries, Adv. Energy Mater. 8 (2018).
[14] X. Li, D. Li, Y. Zhang, P. Lv, Q. Feng, Q. Wei, Encapsulation of enzyme by metal-organic framework for single-enzymatic biofuel cell-based self-powered biosensor, Nano Energy 68 (2020).
[15] L. Luo, S.-H. Chung, A. Manthiram, A trifunctional multi-walled carbon nanotubes/polyethylene glycol (MWCNT/PEG)-coated separator through a layer-by-layer coating strategy for high-energy Li–S batteries, J. Mater. Chem. A 4 (2016) 16805–16811.
[16] K. Kim, P.J. Kim, J.P. Youngblood, V.G. Pol, Surface functionalization of carbon architecture with nano-MnO2 for effective polysulfide confinement in lithium-sulfur batteries, ChemSusChem 11 (2018) 2375–2381.
[17] J. Li, Y. Huang, S. Zhang, W. Jia, X. Wang, Y. Guo, D. Jia, L. Wang, Decoration of silica nanoparticles on polypropylene separator for lithium-sulfur batteries, ACS Appl. Mater. Interfaces 9 (2017) 7499–7504.
[18] N. Mosavati, S.O. Salley, K.Y.S. Ng, Characterization and electrochemical activities of nanostructured transition metal nitrides as cathode materials for lithium sulfur batteries, J. Power Sources 340 (2017) 210–216.
[19] Z. Sun, J. Zhang, L. Yin, G. Hu, R. Fang, H.M. Cheng, F. Li, Conductive porous vanadium nitride/graphene composite as chemical anchor of polysulfides for lithium-sulfur batteries, Nat. Commun. 8 (2017) 14627.
[20] Z. Li, C. Zhou, J. Hua, X. Hong, C. Sun, H.W. Li, X. Xu, L. Mai, Engineering oxygen vacancies in a polysulfide-blocking layer with enhanced catalytic ability, Adv. Mater. 32 (2020) 1907444.
[21] J. Lin, K. Zhang, Z. Zhu, R. Zhang, N. Li, C. Zhao, CoP/C nanocubes-modified separator suppressing polysulfide dissolution for high-rate and stable lithium-sulfur batteries, ACS Appl. Mater. Interfaces 12 (2020) 2497–2504.
[22] T. Foroozan, F.A. Soto, V. Yurkiv, S. Sharifi-Asl, R. Deivanayagam, Z. Huang, R. Rojaee, F. Mashayek, P.B. Balbuena, R. Shahbazian-Yassar, Synergistic effect of graphene oxide for impeding the dendritic plating of Li, Adv. Funct. Mater. 28 (2018).
[23] P.J. Kim, K. Kim, V.G. Pol, Uniform metal-ion flux through interface-modified membrane for highly stable metal batteries, Electrochim. Acta 283 (2018) 517–527.
[24] W. Luo, L. Zhou, K. Fu, Z. Yang, J. Wan, M. Manno, Y. Yao, H. Zhu, B. Yang, L. Hu, A thermally conductive separator for stable Li metal anodes, Nano Lett. 15 (2015) 6149–6154.
[25] J.W. Ming Zhu, Wei-Hong Zhong, Jinle Lan, Gang Sui, Xiaoping Yang, A biobased composite gel polymer electrolyte with functions of lithium dendrites suppressing and manganese ions trapping, Adv. Energy Mater. (2018) 1702561.
[26] P.J.H. Kim, J. Seo, K. Fu, J. Choi, Z. Liu, J. Kwon, L. Hu, U. Paik, Synergistic protective effect of a BN-carbon separator for highly stable lithium sulfur batteries, NPG Asia Mater. 9 (2017) e375.
[27] T. Yang, K. Liu, R. Ren, J. Zhang, X. Zheng, C. Wang, M. Chen, Uniform growth of Li2S promoted by an organophosphorus-based mediator for high rate Li-S batteries, Chem. Eng. J. 381 (2020).
[28] S. Wu, J. Ning, F. Jiang, J. Shi, F. Huang, Ceramic nanoparticle-decorated melt-electrospun PVDF nanofiber membrane with enhanced performance as a lithium-ion battery separator, ACS Omega 4 (2019) 16309–16317.
[29] Y. Cheng, X.Z. Huang, Z.J. Du, J.R. Xiao, S. Zhou, Microstructure and properties of SiC-coated carbon fibers prepared by radio frequency magnetron sputtering Yongshan Wei, Appl. Surf. Sci. 369 (2016) 196–200.
[30] Y.G. Li, H. Yuan, Z.T. Jiang, N. Pan, M.K. Lei, Phase composition and mechanical properties of homostructure NbN nanocomposite coatings deposited by modulated pulsed power magnetron sputtering, Surf. Coat. Technol. 385 (2020).
[31] Z.A. Ghazi, X. He, A.M. Khattak, N.A. Khan, B. Liang, A. Iqbal, J. Wang, H. Sin, L. Li, Z. Tang, MoS_2/Celgard separator as efficient polysulfide barrier for long-life lithium-sulfur batteries, Adv. Mater. 29 (2017).
[32] T.Z. Zhuang, J.Q. Huang, H.J. Peng, L.Y. He, X.B. Cheng, C.M. Chen, Q. Zhang, Rational integration of polypropylene/graphene oxide/Nafion as ternary-layered separator to retard the shuttle of polysulfides for lithium-sulfur batteries, Small 12 (2016) 381–389.
[33] X. Luo, X. Lu, G. Zhou, X. Zhao, Y. Ouyang, X. Zhu, Y.E. Miao, T. Liu, Ion-selective polyamide acid nanofiber separators for high-rate and stable lithium-sulfur batteries, ACS Appl. Mater. Interfaces 10 (2018) 42198–42206.

[34] H.J. Peng, Z.W. Zhang, J.Q. Huang, G. Zhang, J. Xie, W.T. Xu, J.L. Shi, X. Chen, X. B. Cheng, Q. Zhang, A cooperative interface for highly efficient lithium-sulfur batteries, Adv. Mater. 28 (2016) 9551–9558.

[35] J. Wang, Y. Yang, F. Kang, Porous carbon nanofiber paper as an effective interlayer for high-performance lithium-sulfur batteries, Electrochim. Acta 168 (2015) 271–276.

[36] S.H. Kim, J.S. Yeon, R. Kim, K.M. Choi, H.S. Park, A functional separator coated with sulfonated metal–organic framework/Nafion hybrids for Li–S batteries, J. Mater. Chem. A 6 (2018) 24971–24978.

[37] J.H. Lee, J. Kang, S.W. Kim, W. Halim, M.W. Frey, Y.L. Joo, Effective suppression of the polysulfide shuttle effect in lithium-sulfur batteries by implementing rGO-PEDOT:PSS-coated separators via air-controlled electrospray, ACS Omega 3 (2018) 16465–16471.

[38] L. Yin, G. Xu, P. Nie, H. Dou, X. Zhang, MXene debris modified eggshell membrane as separator for high-performance lithium-sulfur batteries, Chem. Eng. J. 352 (2018) 695–703.

[39] X. He, Y. Shuai, L. Na, K. Chen, Y. Zhang, Z. Zhang, F. Gan, High performance lithium-sulfur batteries with facile titanium nitride particles modified separator, Mater. Lett. 215 (2018) 91–94.

[40] M. Liu, H.R. Jiang, Y.X. Ren, D. Zhou, F.Y. Kang, T.S. Zhao, In-situ fabrication of a freestanding acrylate-based hierarchical electrolyte for lithium-sulfur batteries, Electrochim. Acta 213 (2016) 871–878.

[41] C.H. Chang, S.H. Chung, A. Manthiram, Effective stabilization of a high-loading sulfur cathode and a lithium-metal anode in Li-S batteries utilizing SWCNT-modulated separators, Small 12 (2016) 174–179.